

Zurich
Switzerland

Fabio Banfi

Curriculum Vitae

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Experience

Professional

- 04.2024–now **Expert Security Engineer**, *Zühlke Engineering AG*, Zurich, Switzerland
- **Cryptography Consulting:** advised on crypto-agility and post-quantum migration strategies; reviewed cryptographic implementation of cloud split-key encryption and hashing service
 - **Security Consulting:** implemented OpenShift security controls; maintained and automated PKI; supervised external penetration tests; implemented data-loss prevention policies
 - **Blockchain Security:** developed an air-gapped recovery system for digital assets and cryptographic keys
 - **Penetration Testing / Code Review:** performed gray-box assessments and source code analysis of embedded and industrial control devices
 - **Security Testing:** automated IEC 62443-4-2 compliance testing for industrial control devices
- 02–06.2017 **Part-Time Software Engineer**, *University of Zurich*, Zurich, Switzerland
Performed data analysis and visualization in Python
- 04–09.2012 **Software Engineering Internship**, *Siemens Schweiz AG*, Steinhausen, Switzerland
Developed several Windows based network services interfacing with various building automation protocols and related GUIs for configuration in C#

Teaching

- 2018–2022 **Head Teaching Assistant**, *ETH Zurich*, Switzerland
Cryptography Foundations (grad. course), Current Topics in Cryptography (grad. seminar)
- 2017–2022 **Teaching Assistant**, *ETH Zurich*, Switzerland
Algorithms and Probability (undergrad. course), Discrete Mathematics (undergrad. course)

Education

- 2017–2023 **PhD in Computer Science**, *ETH Zurich*, Switzerland
Title: A Composable Treatment of Anonymous Communication
Supervisor: Prof. Ueli Maurer
doi: 10.3929/ethz-b-000637565
- 2015–2017 **Master of Science ETH in Computer Science**, *ETH Zurich*, Switzerland
- 2012–2015 **Bachelor of Science ETH in Computer Science**, *ETH Zurich*, Switzerland

Open Source Projects

- zkgrid.net: a didactic puzzle game that uses zkSNARKs to verify correct solutions
- repartee.xyz: a globally shared word-association game built on the Internet Computer

Computer Skills

- C/C++, Linux, $\LaTeX$, AI-assisted development
- Python, Bash, PowerShell, C#, Java
- Rust, MATLAB, Haskell, Assembly (x86, MIPS, ARM)
- HTML, PHP, CSS, SQL, JavaScript, Markdown

Languages

- Italian *First language*
- English *Fluent in speaking, comprehension, writing and reading*
- German *Fluent in speaking, comprehension, and reading; good in writing*
- French *Fairly good in speaking, comprehension, writing and reading*
- Spanish *Fairly good in comprehension and reading; basic in speaking and writing*

Publications

Christian Badertscher, Fabio Banfi, and Jesus Diaz. What did come out of it? Analysis and improvements of DIDComm messaging. In *ACM SIGSAC Conference on Computer and Communications Security – ACM CCS 2024*. doi:10.1145/3658644.3690300.

Christian Badertscher, Fabio Banfi, and Ueli Maurer. A constructive perspective on signcryption security. In *Security and Cryptography for Networks – SCN 2018*. doi:10.1007/978-3-319-98113-0_6.

Fabio Banfi. SCB mode: Semantically secure length-preserving encryption. In *Fast Software Encryption – FSE 2023*. doi:10.46586/tosc.v2022.i4.1-23.

Fabio Banfi, Konstantin Gegier, Martin Hirt, Ueli Maurer, and Guilherme Rito. Anamorphic encryption, revisited. In *Advances in Cryptology – EUROCRYPT 2024*. doi:10.1007/978-3-031-58723-8_1.

Fabio Banfi and Ueli Maurer. Anonymous authenticated communication. In *Security and Cryptography for Networks – SCN 2022*. doi:10.1007/978-3-031-14791-3_13.

Fabio Banfi and Ueli Maurer. Anonymous symmetric-key communication. In *Security and Cryptography for Networks – SCN 2020*. doi:10.1007/978-3-030-57990-6_23.

Fabio Banfi, Ueli Maurer, Christopher Portmann, and Jiamin Zhu. Composable and finite computational security of quantum message transmission. In *Theory of Cryptography – TCC 2019*. doi:10.1007/978-3-030-36030-6_12.